M2VDet: Midpoints-to-Vertices Detection of Oriented Objects in Remote-Sensing Images

Xueru Xu, *Student Member, IEEE*, Zhong Chen, Guoyou Wang, He Deng, Longji Yu, and Qimeng Chen

***Abstract*— Oriented object detectors provide many scientific solutions for object detection tasks in remote-sensing scenes. Among them, anchor-free oriented object detectors have attracted much attention because of their flexibility and conciseness in recent years. However, the positive sampling strategy adopted by most modern anchor-free oriented detectors is inadequate to reflect remote-sensing objects characterized by arbitrary orientation, densely packed, and extensive scale variation. Besides, there is still a lack of a general representation to describe oriented objects shaped with arbitrary convex quadrilaterals. In this article, we present a one-stage anchor-free oriented object detector, called M2VDet, which focuses on designing the sampling strategy and oriented bounding box (OBB) description. First, an adaptive sampling strategy is proposed to generate positive sample spaces, which take full account of objects' characteristics like scale, shape, and orientation. The resulting candidate regions provide a superior space constraint to construct the heatmap, and then 2-D oriented Gaussian distribution is employed to generate prior labels for positive candidates. Second, a general OBB representation is designed by adopting the midpoints-to-vertices strategy, which provides a unified approach to describe the arbitrary convex quadrilaterals without brutal approximation. Extensive experimental results on four public datasets (e.g., DOTA, HRSC2016, UCAS-AOD, and SODA-A) demonstrate that M2VDet has a simple pipeline, but it can achieve more robust and available performance when compared with state-of-the-art baseline detectors, especially on the detection of densely packed objects with arbitrary orientation as well as small objects.**

***Index Terms*— Anchor-free, oriented bounding box (OBB), oriented object detection, positive sampling strategy, remote sensing.**

I. Introduction

THE continuous progress of universal object detection technology has provided numerous practical solutions for object detection tasks in remote-sensing scenes [1], [2], [3], [4], [5], [6], [7], [8], [9]. Observed from a bird's eye view, remote-sensing images are characterized by arbitrary orientation and are densely packed, which exhibits an explicit gap from natural images. That is, object detectors for natural scenes typically employ axis-aligned bounding boxes to locate objects with upward orientations, which is unbefitting to apply to oriented objects in remote-sensing scenes. Specifically, the horizontal bounding box (HBB) is prone to cover redundant backgrounds of oriented objects, and it fails to locate individual objects in dense arrangement [10]. To this end, modern detection methods aiming at remote-sensing scenes have proposed the oriented bounding box (OBB) for a tighter location. The OBB is more suitable for capturing objects in remote-sensing images [10].

Locating oriented objects is a crucial subtask in oriented object detection. Besides employing the rectangle with a rotation angle to describe the OBB [11], [12], some modern detectors [13], [14], [15], [16], [17], [18], [19] tend to design flexible OBB representations to acquire more precise location information. These detectors utilize the vertices or representative points of the bounding box to describe the OBB. However, the OBB may demonstrate arbitrary shapes in the real scenario, and the existing methods need to convert the OBB to its minimum bounding rectangle when training [14], [15], [16], [17], which may introduce significant deviations in such a rough approximation. Furthermore, failure detection arises for nearly axis-aligned objects under their description [13], [15], [17]. They cope with this issue by predicting nearly axis-aligned objects and other oriented objects through two separate branches, which escape from the intractable problem at the sacrifice of universality. DAFNe [18] and GGHL [19] propose more general representations for arbitrary quadrilaterals, but DAFNe [18] introduces a certain computational cost since it splits the regression process into two phases, and GGHL [19] needs to make different types of approximations for extreme cases. For now, there is still a lack of a general paradigm to describe the OBB shaped with arbitrary quadrilaterals.

In addition, seeing that remote-sensing images are complex and informative, the existing oriented object detectors are inclined to employ a set of predefined anchors for pursuing an outstanding performance [13], [20], [21], [22], [23], [24], [25], [26], [27]. Unfortunately, the orientation information aggravates the design and optimization of dense anchors, resulting in complex, time-consuming computations. Inspired by the anchor-free detectors [4], [5], [6], [7], [8], [9] for the axis-aligned objects, some works focus on the one-stage anchor-free models to detect oriented objects and achieve promising performance [11], [12], [14], [15], [16], [17], [18], [19]. Such anchor-free detectors are independent

Manuscript received 15 November 2022; revised 9 March 2023; accepted 27 August 2023. Date of publication 1 September 2023; date of current version 15 September 2023. *(Corresponding author: Zhong Chen.)*

Xueru Xu, Zhong Chen, Guoyou Wang, Longji Yu, and Qimeng Chen are with the School of Artificial Intelligence and Automation, National Key Laboratory of Multispectral Information Intelligent Processing Technology, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: xuxueru@hust.edu.cn; henpacked@hust.edu.cn; gywang@mail.hust.edu.cn; longji@hust.edu.cn; m202072752@hust.edu.cn).

He Deng is with the School of Computer Science and Technology, Wuhan University of Science and Technology, Wuhan 430081, China (e-mail: denghe@wust.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3310988

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

of priors relevant to datasets and concisely detect objects by key points or dense points, always possessing a faster detection speed. However, anchor-free detectors suffer much from the imbalance of positive and negative samples [28], which causes insufficient supervision information and therefore induces performance degradation. Objects appear with vast scale variation and arbitrary orientation in remote-sensing scenes, which may aggravate sample imbalance and deteriorate detection performance, especially for small objects. Therefore, constructing an adequate positive sample space to split positive and negative locations is beneficial to obtain more performance gains without additional cost.

The existing one-stage anchor-free oriented object detectors mainly inherit the sample assignment strategy from general object detectors. Among them, the dense points-based detectors, as with FCOS [9], always regard all locations inside the bounding box as positive samples, which may misallocate some negative samples as positive samples for those objects with complex geometry. By contrast, the keypoints-based detectors define compact positive sample spaces located inside the OBB, typified by O^2-DNet [15], BBAVectors [17], and GGHL [19]. To be specific, predefined metrics such as the intersection over union (IoU) thresholds are adopted to determine the scope of the positive sample spaces. However, such static metrics are not optimal for objects with different sizes and shapes, resulting in rare positive samples for small objects when selecting a higher threshold or poor-quality positive samples for large objects when selecting a lower threshold [29]. Considering that objects' scales in remote-sensing scenes vary shapely, sample assignment strategies should be adaptable enough to take full consideration of the objects' scale, shape, and orientation at the same time.

Since the pipeline of keypoints-based detectors is conceptually simple, we adopt their core thoughts to propose a simple, yet effective oriented object detector for remote-sensing objects. We present an adaptive sampling strategy to generate a flexible positive candidate region and a general representation to describe the OBB shaped with the arbitrary quadrilateral. Our contributions are summarized as follows.

1) An effective one-stage anchor-free oriented object detector is presented to detect remote-sensing objects with arbitrary orientations and diverse scales. This detector can achieve competitive performance merely via designing an adaptive sampling strategy and a general OBB representation method without any support from complicated CNN structures.
2) An adaptive sampling strategy is proposed to generate positive candidate regions by sufficiently considering the objects' characteristics like shape, scale, and orientation. Then, 2-D-oriented Gaussian distribution is utilized for modeling the positive candidates, based on which prior labels are assigned for each location inside the candidates. The generated Gaussian candidates contribute more to detecting oriented objects shaped with arbitrary orientations and diverse scales in remote-sensing images.
3) A general OBB representation method with the "midpoints-to-vertices" strategy is designed to describe the OBB. The prediction of the OBB is thus converted to two steps: obtaining the midpoints of its circumscribed HBB's edges first and then predicting the decomposed vectors from the midpoints to the OBB's vertices. The novel representation can detect nearly axis-aligned objects and oriented objects via a unified regression branch for its credible capability to describe arbitrary convex quadrilaterals.

The remainder of our article is organized as follows: In Section II, we review some related works and analyze the remaining challenge. In Section III, we illustrate our approach elaborately. We place our experimental details in Section IV, and the results are analyzed comprehensively. In Section V, we make a brief conclusion to this article.

II. Related Works

A. Anchor-Free Oriented Object Detectors

Oriented object detectors generally derive from the typical paradigm of axis-aligned object detectors and are divided into anchor-based detectors [13], [20], [21], [22], [23], [24], [25], [26], [27] and anchor-free detectors [11], [12], [14], [15], [16], [17], [18], [19]. Since remote-sensing images are characterized by complex backgrounds, extensive scales, and arbitrary directions, anchor-based detectors always manifest outstanding performance by employing specific anchor priors, resulting in a mature application in remote-sensing scenes. However, the predefined anchors of OBBs are more intricate in design and optimization. Inspired by single-stage anchor-free detectors in the horizontal object detection tasks [4], [5], [6], [7], [8], [9], anchor-free detectors of oriented objects come into focus. These kinds of models remove a mass of handcrafted parameters about anchors, thus making the detection process concise.

Consistent with classical anchor-free detectors in the horizontal object detection task, anchor-free detectors for oriented objects can be roughly divided into keypoints-based detectors [12], [15], [16], [17], [19] and dense points-based detectors [11], [14], [18]. Keypoints-based detectors are prone to employ CenterNet [6] and RepPoints [8] as the baseline architecture, while dense points-based detectors always utilize the detection architecture of FCOS [9]. To predict oriented objects, prime architectures are equipped with a new regression branch to predict orientation information. IENet [11] is the first anchor-free oriented object detector based on FCOS, and it designs an additional branch to predict the rotation parameters of the OBB. After that, oriented object detectors [14], [15], [16], [17], [18], [19] directly substitute the original regression branch to the OBB regression branch, which makes the detection head more streamlined.

B. Sample Assignment Strategy

Sample assignment is critical when training anchor-free oriented detectors, which leaves a notable impact on detection performance. Modern anchor-free oriented detectors generally adopt center sampling strategies, selecting a specific region inside the OBB as the positive sample space. O^2-DNet [15], P-RSDet [16], and BBAVectors [17] develop circular regions around the center point following most HBB detectors, but a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

circular region can hardly represent the shape and orientation of the OBB. To solve this problem, GGHL [19] further employs a rotating elliptical region to describe positive sample space, which is more in line with the geometrical shape of the OBB. Among them, the scope of positive sample spaces is determined through static metrics, typified by a predefined IoU threshold. However, selecting an optimal static metric for objects with diverse shapes and scales remains challenging.

In practice, several dynamic assignment strategies have been proposed for generic object detection tasks. ATSS [28] adaptively attains the IoU threshold based on the statistical characteristic of candidate anchors. FreeAnchor [30], AutoAssign [31], and PAA [32] allow objects to dynamically select positive and negative samples when training according to prediction results of predefined anchors. However, most of these methods are designed based on anchors and rarely pay attention to the shape information of objects. Inspired by ATSS [28], Hou et al. [33] propose flexible shape-adaptive selection and shape-adaptive measurement strategies for the oriented object detection task, which can select positive samples according to the shape and characteristics distribution of oriented objects. In view that objects' scale varies immensely in remote-sensing scenes, we attempt to construct the adaptive positive sample spaces by considering the objects' scale, shape, and orientation details for a better balance of positive and negative samples when training.

C. OBB Representation

As a critical content of oriented object detector, the current OBB representation methods mainly include two kinds: angle-based and coordinate-based. The angle-based representation can hardly describe the arbitrary quadrilateral because it approximates the OBB as its rotated rectangle. Furthermore, this representation suffers from the inherent periodicity issue of the angle, as analyzed in GWD [34]. Existing detectors have proposed several practical solutions to address this problem, mainly by optimizing the loss function of the regression branch, such as modulated loss [35], PIoU [36], GWD [34], KLD [37], and KFIoU [38]. In contrast, the coordinate-based method provides a more intuitive method by describing OBB with the coordinates of its vertices or representative points. Besides directly predicting the vertex coordinates, Gliding Vertex [13] expresses the OBB as the slip distance from four vertices of the circumscribed HBB to the corresponding four vertices of the OBB along the HBB's edge. Oriented RCNN [26] further replaces vertices of the circumscribed HBB with midpoints of the HBB's edge. For the case where the vertices of an arbitrary convex quadrilateral may not entirely fall on the circumscribed HBB, GGHL [19] proposes a simple approximation method to preprocess the extraordinary OBB. DAFNe [18] predicts the center point first and then regresses the vertices from the predicted center with two separate branches. In addition, some detectors design novel representations instead of regressing four vertices. Axis-Learning [14] predicts the central axis of the OBB and the length of the OBB's side which is vertical to the axis; O^2-DNet [15] detects the OBB by a pair of middle lines; BBAVectors [17] uses the boundary-aware vectors from the center point to four edges of the OBB to capture the OBB.

Compared with the angle-based representation, the coordinate-based representation is more flexible and accessible to describe arbitrarily convex quadrilaterals in nature. However, there still exists the boundary effect that results in failure detection for nearly axis-aligned objects. Gliding Vertex [13], O^2-DNet [15], and BBAVectors [17] all indicate this issue and propose two separate branches to regress nearly axis-aligned objects and other oriented objects. This solution effectively averts the boundary problem, yet somewhat sacrifices the location accuracy. Moreover, most detectors cope with OBB as its minimum boundary rectangle to simplify the optimization procedure, which presents an obstacle to predicting the arbitrary convex quadrilaterals. Therefore, how to design a general representation to predict the OBB shaped with the arbitrary convex quadrilateral remains to be explored.

III. Proposed M2VDet Framework

The framework of the proposed M2VDet is shown in Fig. 1. As a one-stage anchor-free oriented object detector, we use ResNet as the backbone, followed by a feature fusion module (FFM) to transfer the high-level semantic information to the low-level details, then feed the single high-resolution feature map into the detection head for object prediction. The detection head includes three branches: the heatmap prediction branch, the OBB regression branch, and the center point offset branch. The heatmap prediction branch is utilized to detect the positive location and corresponding category confidence at each location, while the OBB regression branch and center point offset branch are used to regress OBB at each positive location. For an input image with the height H_0 and the width W_0, the FFM outputs a feature map with the size of $C \times (H_0/s) \times (W_0/s)$, where s refers to the downsampling rate and C is the channel number. We set $s = 4$ and $C = 256$ in this work. Then the output is fed into the detection head for obtaining the heatmap, OBB parameters, and center point offset with the size of $K \times (H_0/s) \times (W_0/s)$, $20 \times (H_0/s) \times (W_0/s)$, and $2 \times (H_0/s) \times (W_0/s)$, where K is the number of categories.

Objects' scale varies sharply in remote-sensing scenes, which makes anchor-free detectors suffer much from the imbalance of positive and negative samples. To this end, we first propose an adaptive sampling strategy for generating positive sample spaces in line with objects' shape, size, and orientation simultaneously. The label assignment of the sample space is acquired through 2-D-oriented Gaussian distribution modeling. Besides, in response to the lack of a unified representation to describe oriented objects shaped with arbitrary quadrilaterals, we propose a general OBB representation referring to the "midpoints-to-vertices" strategy. To be specific, at each positive location, the circumscribed HBB is first obtained by predicting the offset from the sample location to the center point and the HBB's size at the center point, then four vertices of the OBB can be described as the decomposed midpoints-aware vectors from midpoints of the HBB's edge to four vertices of the OBB. Such a novel OBB representation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

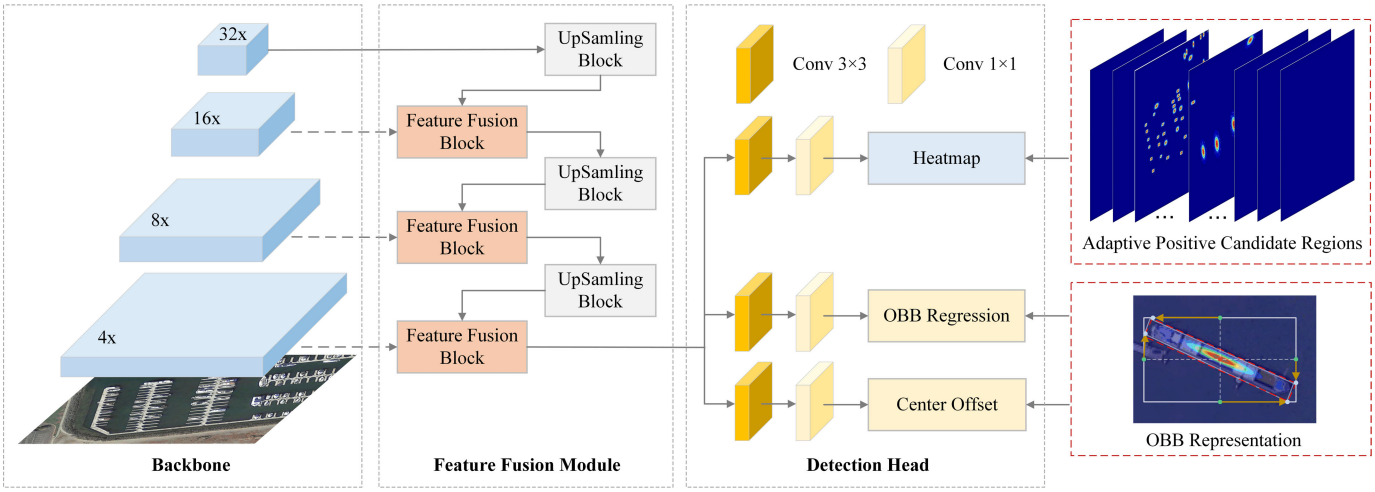


Fig. 1. M2VDet framework. The architecture is the U-shape network with skip connections to fusion features across scales, and the single-scale feature map with 4× downsampling rate is fed into the detection head to conduct OBB regression and classification. The upsampling block is the bilinear interpolation operation followed by a 3 × 3 convolutional layer. Then, the upsampling feature map is concatenated with the shallow one from the backbone in the channel dimension and a 1 × 1 convolutional layer is utilized for channel adjustment, which is the main process of the feature fusion block. The detection head contains three subnetworks: the heatmap branch produces one channel heatmap per category and the heatmap value is directly regarded as the confidence score; the OBB regression branch is utilized to predict the OBB at each positive location, and the center offset branch acts as the compensation for numerical precision. The network is a fully convolutional architecture and is conceptually simple.

provides a unified pattern to regress nearly axis-aligned objects and oriented objects in the same prediction branch.

A. Adaptive Sampling Strategy

Anchor-based detectors usually define positive or negative anchors based on the IoU between anchor boxes and their ground truth. By contrast, anchor-free object detectors develop different strategies to identify the positive samples, that is, build a candidate region co-centered with the object and get its heatmap with Gaussian-like or Centerness-like weighting mechanisms. Note that the scope of the candidate region satisfies some predefined constraints such as the IoU threshold. CenterNet [6] employs a standard Gaussian distribution to model positive candidates of each object, then the circular Gaussian heatmap is obtained to detect the center point, and the HBB regression is conducted only at Gaussian peak points. Toward oriented object detection tasks, BBAVectors [17] and DRN [12] adopt the same approach as the CenterNet [6] to generate the Gaussian heatmap, whereas the circular Gaussian region cannot convey geometric shape information of oriented objects. In contrast, GGHL [19] utilizes the oriented elliptical Gaussian regions to present the positive candidates and treats all locations inside Gaussian regions as positive samples, which are more suitable for oriented objects. However, the scale of candidate regions is still determined through the predefined IoU threshold, which may be suboptimal for variant scales of objects.

We propose a feasible method to generate positive candidate regions according to objects' scale and shape. We first develop the positive candidate region co-centered and co-axial with the OBB, whose scope is dynamically determined by the object's size and shape. Then, we employ the 2-D-oriented Gaussian distribution to model the candidates as an oriented elliptical region and get label assignments at each location according to the Gaussian probability density function. The implementation details are as follows.

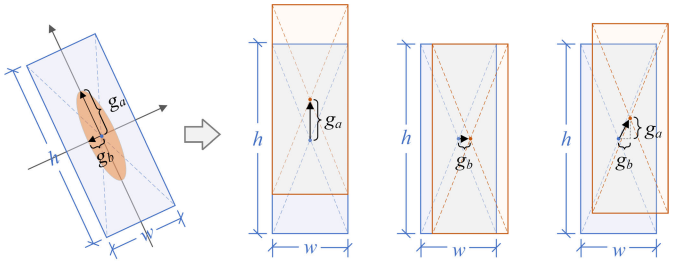


Fig. 2. Scope of the positive candidate region, w and h represent the width and height of the OBB's minimum bounding rectangle, respectively, g_a and g_b indicate the radius of the candidate region.

1) For each OBB with an arbitrary shape in the feature map, we suppose there is a smaller region co-centered with the original OBB for the positive sample space. The locations inside the sample space need to satisfy the condition that there exists a new bounding box with the same size whose IoU with the original OBB is larger than the IoU threshold t. As shown in Fig. 2, the IoU between the blue bounding box and the orange bounding box is greater than t.
When implementing, we first obtain the minimum bounding rectangle of the initial OBB and describe it as $[x, y, w, h, \theta]$, then the long-axis and short-axis of the candidate region need to meet the following conditions:

$$\frac{g_a}{\max(h,w)} \le \frac{1-t}{1+t} \tag{1a}$$

$$\frac{g_b}{\min(h,w)} \le \frac{1-t}{1+t} \tag{1b}$$

$$\frac{g_a}{\max(h,w)} + \frac{g_b}{\min(h,w)} - \frac{g_a \times g_b}{h \times w} \le \frac{1-t}{1+t} \tag{1c}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

where g_a and g_b form a new region for positive candidates, g_a and g_b represent the semimajor axis and the semiminor axis, respectively. In our work, we set

$$g_a = \sqrt{\frac{1-t}{1+t}} \times \max(h, w) \tag{2a}$$

$$g_b = \sqrt{\frac{1-t}{1+t}} \times \min(h, w). \tag{2b}$$

We describe the positive candidate region as $P[x, y, g_a, g_b, \theta]$, which is a rotated elliptical region co-centered and co-axial with the OBB. Compared to the standard circular region, the rotated elliptical region considers more facts about the objects' shape and orientation, which is more practical for oriented objects.

2) We use the IoU matching strategy, consistent with the mainstream approach, to obtain the scope of positive candidate regions. But different from most methods that adopt a constant threshold, like 0.3 or 0.7, we assign an adaptive threshold for each object according to its size. Primarily, we first normalize the OBB's short side by dividing the size of the feature map, then we conduct a monotonic increasing function to map the normalized value into (0.5, 1). The mapping value is regarded as the IoU threshold concisely. The mapping function is

$$k = \left(\min(w, h)/S_f\right)^r \tag{3a}$$

$$t = \frac{1}{1+e^{-k}} \tag{3b}$$

where S_f represents the size of the feature map. r is a hyperparameter that controls the magnitude of normalization. t represents the IoU thresh that can adaptively adjust according to the scale of the OBB.
Fig. 3 indicates the mapping function curve of the adaptive IoU threshold related to the normalized short side of the OBB. We set $r \in (0, 1]$ to guarantee that the adaptive threshold is more sensitive to small scales. It is observed that the smaller value of r results in a relatively compact distribution of the adaptive IoU threshold, which may be inconducive to preserving the adaptability of the IoU threshold to the scale. As the value of r increases, the distribution range of the IoU threshold becomes slacker. Meanwhile, the IoU threshold is more sensitive to large-scale objects and less sensitive to small objects. Therefore, there exist one or more values that not only maintain the threshold's adaptability to the scale, but also balance the threshold's sensitivity to cross-scale objects.

3) We employ a general 2-D Gaussian distribution to model the positive candidate region $P[x, y, g_a, g_b, \theta]$. Suppose the positive candidate region submits to the 2-D Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ represent the mean vector and the covariance matrix of two dimensions, respectively. Associated with geometrical transformation, $\boldsymbol{\mu}$ controls the spatial translation, while $\boldsymbol{\Sigma}$ contributes to the spatial rotation and scaling. The real symmetric matrix $\boldsymbol{\Sigma}$ is orthogonally diagonalized and decomposed into

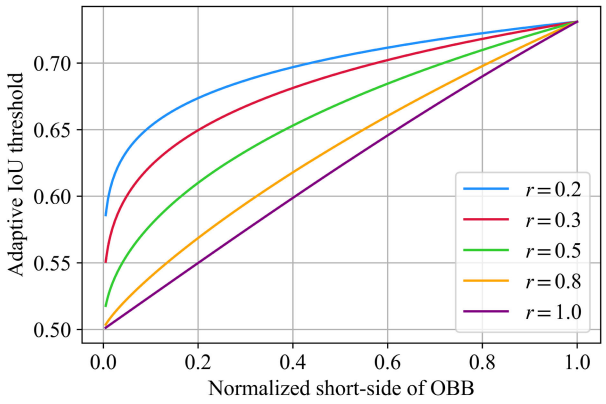


Fig. 3. Mapping function curve of the adaptive IoU threshold related to the normalized short side of the OBB.

$$\boldsymbol{\Sigma} = \boldsymbol{R}\boldsymbol{\Lambda}\boldsymbol{R}^{\mathrm{T}} \tag{4}$$

$$\boldsymbol{R} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \quad \boldsymbol{\Lambda} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} {g_a}^2 & 0 \\ 0 & {g_b}^2 \end{bmatrix} \tag{5}$$

where $\boldsymbol{R}$ is a real orthogonal matrix to represent the rotation matrix and $\boldsymbol{\Lambda}$ is a diagonal matrix of the eigenvalues indicating the scaling information. θ is the rotation angle, $\theta \in [0, \pi)$. λ_1 and λ_2 are the eigenvalues of the diagonal matrix $\boldsymbol{\Lambda}$, $\lambda_1 = {g_a}^2$, and $\lambda_2 = {g_b}^2$.

We then assign weights for the whole positive candidate region through the Gaussian probability density function. For $\boldsymbol{X} = [x_1, x_2]^{\mathrm{T}} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the probability density function of 2-D Gaussian distribution is defined as

$$f(\boldsymbol{X}) = \frac{1}{2\pi \times |\boldsymbol{\Sigma}|^{1/2}} e^{-\frac{1}{2}[(X-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(X-\boldsymbol{\mu})]}. \tag{6}$$

For each point $[x, y] \in P$, we generate the prior probability through the normalized Gaussian value, and then we can obtain the heatmap ground truth. As shown in Fig. 4, the heatmap ground truth is described as $G \in \mathbb{R}^{H \times W \times K}$, where K is the number of categories. We generate a heatmap for each category to predict positive locations and corresponding confidence. The predicted heatmap value is regarded as the confidence score belonging to each category.

B. General OBB Representation

Unlike assuming the Gaussian peak points as positive locations in CenterNet [6], we set the position inside the Gaussian candidate region as the positive location if its Gaussian probability value is greater than a threshold δ, and the OBB regression subtask is then carried out at all positive locations. Such a simple selection can reduce the influence of accurately extracting the keypoints from the heatmap on the OBB regression results. Given that objects are generally shaped with arbitrary convex quadrilaterals, we propose a "midpoints-to-vertices" strategy to describe the OBB at each positive location. To be specific, we utilize its circumscribed HBB and a set of midpoints-aware vectors for a general representation, as shown in Fig. 5. The details are as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

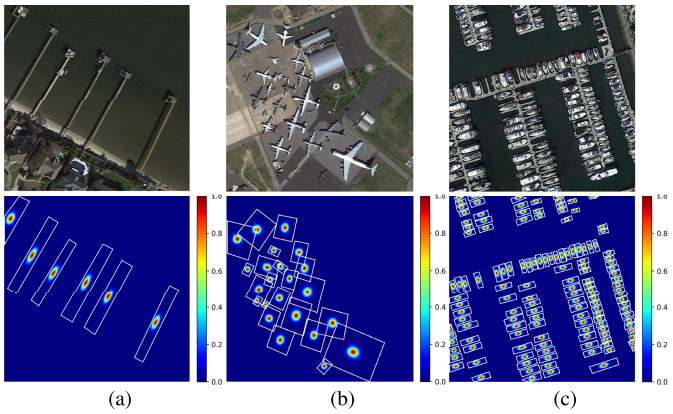
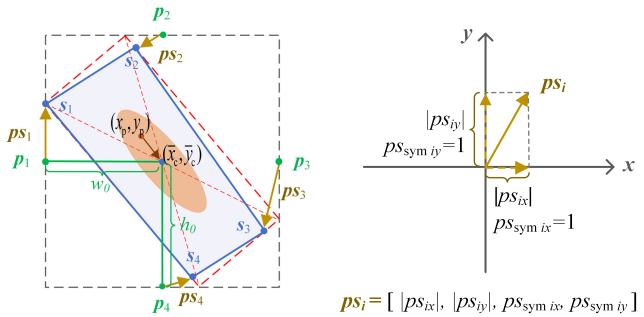


Fig. 4. Heatmaps ground truth for objects. (Top row) Original remote-sensing images. (Bottom row) Corresponding heatmaps are displayed. (a) Large-scale objects. (b) Median-scale objects. (c) Small-scale objects.



(a) (b)

Fig. 5. Proposed general OBB representation. (a) OBB is described as its circumscribed HBB and the corresponding midpoints-aware vectors from midpoints of the HBB's edge to vertices of the OBB. (b) Midpoints-aware vectors are decomposed along the axes into the amplitude components and the direction components.

1) We employ OBB's minimum boundary rectangle as an approximation to simplify the description of its corresponding circumscribed HBB. In detail, we obtain the circumscribed HBB from its minimum boundary rectangle first, then we use $\boldsymbol{H}_{x_p,y_p} = [c_x, c_y, w_0, h_0]$ to describe the circumscribed HBB at the location $[x_p, y_p]$, where c_x and c_y represent the offset from the current location $[x_p, y_p]$ to the quantified center point $[\bar{x}_c, \bar{y}_c]$, w_0 and h_0 are half-width and half-height of the circumscribed HBB, respectively. Inspired by DAFNe [18], we introduce the center point of the HBB to predict the offset from the other positive location to the center point first, and as a result, we can utilize only two parameters, w_0 and h_0, to describe the HBB's size at all positive locations. Note that we retain the center point offset branch like CenterNet [6] to compensate for the accuracy degradation during the downsampling process. Based on the circumscribed HBB, we can further calculate the coordinate of midpoints $\boldsymbol{p}_1$, $\boldsymbol{p}_2$, $\boldsymbol{p}_3$, and $\boldsymbol{p}_4$ of the left, top, right, and bottom edges

$$\boldsymbol{p}_1 = \left(x_p + c_x - w_0,\ y_p + c_y\right) \tag{7a}$$
$$\boldsymbol{p}_2 = \left(x_p + c_x,\ y_p + c_y - h_0\right) \tag{7b}$$
$$\boldsymbol{p}_3 = \left(x_p + c_x + w_0,\ y_p + c_y\right) \tag{7c}$$
$$\boldsymbol{p}_4 = \left(x_p + c_x,\ y_p + c_y + h_0\right). \tag{7d}$$

2) We set the vertex with the smallest x-coordinate value as the start point and sort four vertices of the OBB clockwise, and then the one-to-one correspondence is established between the ordered vertices and the four midpoints. Subsequently, we use midpoints-aware vectors $\boldsymbol{ps}_1$, $\boldsymbol{ps}_2$, $\boldsymbol{ps}_3$, and $\boldsymbol{ps}_4$ to describe the OBB, which means the vectors from the midpoints $\boldsymbol{p}_1$, $\boldsymbol{p}_2$, $\boldsymbol{p}_3$, and $\boldsymbol{p}_4$ of the circumscribed HBB's edges to the ordered OBB vertices $\boldsymbol{s}_1$, $\boldsymbol{s}_2$, $\boldsymbol{s}_3$, and $\boldsymbol{s}_4$, respectively. Note that $\boldsymbol{ps}_i = [ps_{ix}, ps_{iy}]$ is a 1×2 vector normalized by dividing the w_0 and h_0 for a relatively concentrated value distribution. Our design for OBB representation can cope with the extreme case where vertices do not fall on the edge of circumscribed HBB, and midpoint-aware vectors can express arbitrary OBBs.
3) We observe that the boundary effect still exists on midpoints-aware vectors, manifested as failure detection when the object is nearly axis-aligned. To solve this issue, we further decompose the ith midpoints-aware vector $\boldsymbol{ps}_i$ into the amplitude components $|ps_{ix}|$ and $|ps_{iy}|$ and the direction components $ps_{\text{sym }ix}$ and $ps_{\text{sym }iy}$ along the axes. The amplitude component only represents the length between midpoints to its corresponding vertices along the axes, satisfying the constraints $|ps_{ix}|, |ps_{iy}| \geq 0$. The direction component indicates the direction from midpoints to its corresponding vertices along the axes, which is the discrete value with $ps_{\text{sym }ix}, ps_{\text{sym }iy} \in \{-1, 1\}$. We set

$$ps_{\text{sym }i[x,y]} = \begin{cases} 1, & \text{if } ps_{i[x,y]} \geq 0 \\ -1, & \text{if } ps_{i[x,y]} < 0. \end{cases} \tag{8}$$

Thus, a 1×16-dimensional vector $\boldsymbol{R}_{x_p,y_p}$ is used to describe OBB vertices at the location $[x_p, y_p]$

$$\begin{aligned} \boldsymbol{R}_{x_p,y_p} &= [\boldsymbol{ps}_1, \boldsymbol{ps}_2, \boldsymbol{ps}_3, \boldsymbol{ps}_4] \\ &= [|ps_{ix}|, |ps_{iy}|, ps_{\text{sym }ix}, ps_{\text{sym }iy}] \end{aligned} \tag{9}$$

where $i = 1, 2, 3, 4$.

We further conduct exponent arithmetic to transform values of the amplitude component greater than zero and expand the numerical distribution range by multiplying a constant for better regression

$$|ps_{ix}| \leftarrow d_s \times e^{|ps_{ix}|} \tag{10a}$$
$$|ps_{iy}| \leftarrow d_s \times e^{|ps_{iy}|} \tag{10b}$$
$$ps_{\text{sym }ix} \leftarrow d_s \times ps_{\text{sym }ix} \tag{10c}$$
$$ps_{\text{sym }iy} \leftarrow d_s \times ps_{\text{sym }iy}. \tag{10d}$$

In our article, we set d_s equal to the downsampling rate of the feature map, $d_s = 4$.

To sum up, the arbitrarily OBB at $[x_p, y_p]$ is defined as $\boldsymbol{O}_{x_p,y_p} = [\boldsymbol{H}_{x_p,y_p}, \boldsymbol{R}_{x_p,y_p}]$, where $\boldsymbol{H}_{x_p,y_p}$ contains four parameters related to the circumscribed HBB and $\boldsymbol{R}_{x_p,y_p}$ contains 16 parameters to describe the transformation from midpoints of the HBB's edge to the corresponding vertices of the OBB. We finally obtain a 1×20-dimensional vector at each positive location $[x_p, y_p]$ to predict the OBB. Designing a separate branch for axis-aligned object detection is no longer necessary under our representation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

C. Loss Function

The training process is performed by minimizing the joint loss function

$$L = \alpha_1 \times L_h + \alpha_2 \times L_{\text{gwd}} + \alpha_3 \times L_o \tag{11}$$

where L_h is the heatmap loss, L_{gwd} is the regression loss of the circumscribed HBB, and L_o is the regression loss of the midpoints-aware vectors. L_{gwd} and L_o are merely calculated on positive locations. α_1, α_2, and α_3 are balance weightings for each loss item. We set $\alpha_1 = 1$, $\alpha_2 = 5$, and $\alpha_3 = 0.3$ in this work.

1) Heatmap Loss: Our article employs the heatmap to detect the positive locations and regards the predicted heatmap values as the category confidence for the classification subtask. As with CornerNet [4], we consider the Gaussian peak point inside the Gaussian candidate region as the positive location. For a better balance of positive and negative samples when training, all negative locations inside the Gaussian candidates are penalized by the predicted heatmap value. We follow the variant focal loss to train the heatmap:

$$L_h = -\frac{1}{N}\sum_i \begin{cases} (1-p_i)^{\gamma}\log(p_i), & \text{if } \hat{p}_i = 1 \\ (1-\hat{p}_i)^{\beta} p_i{}^{\gamma}\log(1-p_i), & \text{otherwise} \end{cases} \tag{12}$$

where $\hat{p}_i$ and p_i indicate the ground-truth heatmap value and the predicted heatmap value at location i, and N is the number of objects. γ is a hyperparameter that balances easy or hard examples, and β is a hyperparameter that controls the penalty of the negative locations inside the Gaussian candidates. We set $\gamma = 2$ and $\beta = 4$ empirically.

2) OBB Regression Loss: GWD [34] systematically analyzes the inconsistency between the *l*n-based loss and IoU-based metric in the OBB regression process and proposes a novel regression loss based on Gaussian Wasserstein distance for the rotated bounding box. GWD-based loss is an approximate and differentiable IoU-induced loss that fits more with the IoU-based metric. We describe OBB as a combination of the circumscribed HBB and the decomposed midpoints-aware vectors, thus we set different loss functions against two components for robust supervision.

We suppose that the circumscribed HBB $[x, y, w, h]$ and its corresponding ground truth $[\hat{x}, \hat{y}, \hat{w}, \hat{h}]$ are subject to 2-D Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathcal{N}(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}})$, respectively. Thus, we measure the similarity of two distributions via the Gaussian–Wasserstein distance

$$\begin{aligned} d^2 &= \left\|\boldsymbol{\mu} - \hat{\boldsymbol{\mu}}\right\|_2^2 + \left\|\boldsymbol{\Sigma}^{1/2} - \hat{\boldsymbol{\Sigma}}^{1/2}\right\|_F^2 \\ &= (x-\hat{x})^2 + (y-\hat{y})^2 + \frac{(w-\hat{w})^2 + (h-\hat{h})^2}{4} \\ &= l_2 - \text{norm}\left(\left[x, y, \frac{w}{2}, \frac{h}{2}\right]^{\mathrm{T}}, \left[\hat{x}, \hat{y}, \frac{\hat{w}}{2}, \frac{\hat{h}}{2}\right]^{\mathrm{T}}\right) \end{aligned} \tag{13}$$

where d^2 is Gaussian–Wasserstein distance, which is approximately equivalent to the l_2-norm for horizontal boxes, $[x, y]$ and $[\hat{x}, \hat{y}]$ are the center points of the predicted HBB and the ground-truth HBB, respectively, and $[w, h]$ and $[\hat{w}, \hat{h}]$ are the width and height of the predicted HBB and the ground truth HBB.

We then perform a nonlinear transformation and convert GWD into a standard IoU-based loss form

$$f\left(d^2\right) = 1 - \frac{1}{2+\sqrt{d^2}}. \tag{14}$$

In our article, we combine the OBB regression branch and center offset branch to supervise the regression of the circumscribed HBB

$$\begin{aligned} d_p^2 = &\left(c_x + o_x - \hat{c}_x - \hat{o}_x\right)^2 + \left(c_y + o_y - \hat{c}_y - \hat{o}_y\right)^2 \\ &+ (w_0 - \hat{w}_0)^2 + (h_0 - \hat{h}_0)^2 \end{aligned} \tag{15}$$

$$L_{\text{gwd}} = \frac{1}{K}\sum_{p=1}^{K}\left(1 - \frac{1}{2+\sqrt{d_p^2}}\right) \tag{16}$$

where $[o_x, o_y]$ and $[\hat{o}_x, \hat{o}_y]$ are the center point offset of the predicted value and the ground truth. $[c_x, c_y, w_0, h_0]$ and $[\hat{c}_x, \hat{c}_y, \hat{w}_0, \hat{h}_0]$ represent the predicted HBB and the corresponding ground truth. d_p^2 indicates the GWD of the p location. K is the number of positive locations for the regression branch.

For decomposed midpoints-aware vectors $\boldsymbol{R}_{x_p, y_p}$, we employ smooth $L1$-loss to optimize

$$L_o = \frac{1}{K}\sum_{p=1}^{K} \text{smooth } L1(\boldsymbol{R}_p - \hat{\boldsymbol{R}}_p) \tag{17}$$

$$\text{smooth } L1(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{18}$$

where $\boldsymbol{R}_p$ and $\hat{\boldsymbol{R}}_p$ represent the predicted value and ground-truth value at p location, respectively.

IV. Experiments

A. Datasets

We conduct experiments on four datasets: DOTA1.0 [10], HRSC2016 [39], UCAS-AOD [40], and SODA-A [41]. The instances of these datasets are annotated with OBBs.

1) *DOTA1.0:* The DOTA1.0 dataset contains 2806 aerial images with an extensive resolution ranging from 800 × 800 to 4000 × 4000. The training set has 1411 images, while the validation set has 458 images. Objects in the training and validation sets are all annotated by arbitrary quadrilaterals and are classified into 15 categories. We use abbreviations to represent each category: plane (PL), baseball diamond (BD), bridge (BR), ground field track (GFT), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). The testing set contains 937 images without annotations. Following most of the previous works, the training set and validation set are employed to train the proposed model, and the testing set is used to evaluate the performance. We use multiscale cropping (MSC) with the scales (0.5, 1.0) and then crop the raw image into patches of 600 × 600 pixels with an overlap of 100 pixels.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

2) *HRSC2016:* HRSC2016 is a ship detection dataset collected from several famous harbors. It contains 1061 aerial images with the size ranges from 300 × 300 to 1500 × 900. The training, validation, and testing sets include 436, 181, and 444 images. We train the model on the training set, validate the model on the validation set, and evaluate the performance on the testing set. All images are resized to 800 × 800 pixels without distortion when training and testing.
3) *UCAS-AOD:* The UCAS-AOD dataset is designed for airplane and car detection. It contains 1510 images and 14 596 instances, including 1000 images with 7482 airplanes and 510 images with 7114 cars. Following the former conventions, we randomly select 50% for training, 20% for validation, and 30% for testing. Then we train the model on the training set and evaluate the performance on the testing set. All images are resized to 800 × 800 pixels without distortion when training and testing.
4) *SODA-A:* The SODA-A dataset is specified for detecting oriented objects with small scales under aerial scenarios. It includes 2513 aerial images with a high resolution of 4700 × 2700 pixels and possesses 872 069 instances with significant density diversity. The image set is separated into the train, validation, and test sets with approximate percentages of 40%, 25%, and 35%. All instances are classified into nine categories: airplane, helicopter, small vehicle, large vehicle, ship, container, storage tank, swimming pool, and windmill. Note that instances with areas larger than 2000 pixels are discarded for exclusively detecting small objects. Following [41], we crop the raw images into 800 × 800 patches with an overlap of 150 pixels. We train the model on the train set, validate the model on the validation set, and evaluate the performance on the test set.

B. Implementation Details

We first pretrain the model on the DOTA1.0 dataset with the input resolution of 608 × 608 on the training set and evaluate the performance on the validation set, employing the ImageNet pretrained backbone. The training process is not broken up until the joint loss of the validation set converges. Then we fine-tune the pretrained weight to the DOTA1.0 dataset, HRSC2016 dataset, and UCAS-AOD dataset with the input resolution of 800 × 800 and train the model for 50, 50, and 30 epochs, and the batch size is set to 16, 2, and 8, respectively. For the SODA-A dataset, we employ the ImageNet pretrained backbone and first conduct the experiment with the input resolution of 800 × 800 for 80 epochs, and the batch size is set to 8. In accordance with the experimental details presented in [41], we further resize image patches to 1200 × 1200 during training and testing to retain more information for small objects. To make the training process more efficient, we leverage the model weights trained at the input resolution of 800 × 800 to initialize the model at 1200 × 1200 resolution. Specifically, we employ the weight of the 40th epoch at 800 × 800 input resolution for initialization and continue training with a higher input resolution of 1200 × 1200 for another 50 epochs, and the batch size is set to 4. For all experiments in our article, Adam is used as the optimizer with the initial learning rate of 0.00012, and the exponential reducing strategy is employed to decay the learning rate during training. We apply random flipping, random rotating, random cropping, and color jittering as data augmentation during the involved training process.

In the inference stage, we first apply a 3 × 3 max-pooling layer on the predicted heatmaps to exact the points with maximum heatmap values. Then, we set the confidence threshold to 0.15 to obtain the positive locations whose heatmap values are greater than the confidence threshold, and OBB parameters and center point offsets at these positive locations are further involved in decoding OBB's vertices. Finally, a rotated nonmaximum suppression (NMS) with a threshold of 0.2 is applied to the decoded OBB for postprocessing. Note that the DOTA1.0 dataset adopts two scales (0.5, 1.0) for multiscale testing, which is the same as the training process.

For the DOTA, HRSC2016, and UCAS-AOD datasets, we employ the mAP metric at the IoU threshold of 0.5. For the SODA-A dataset, we adopt the evaluation metrics proposed in [41], which is a COCO-style protocol, including AP, AP_{50}, AP_{75}, AP_{eS}, AP_{rS}, AP_{gS}, and AP_N. Especially, AP refers to the averaged mAP with the IoU threshold varying from 0.5 to 0.95 at an interval of 0.05. The mAP at the IoU threshold of 0.5 and 0.75 are identified as AP_{50} and AP_{75}, respectively. Moreover, AP_{eS}, AP_{rS}, AP_{gS}, and AP_N are scale-dependent evaluation metrics for *extremely small* (0–144 pixels), *relatively small* (144–400 pixels), *generally small* (400–1024 pixels), *normal* (1024–2000 pixels) objects. All experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU.

C. Comparison With the State-of-the-Art

1) Results on DOTA1.0: We first compare the performance of the proposed method with several one-stage anchor-free detectors on the DOTA1.0 dataset, as shown in Table I. With ResNet-101 as the backbone, we can achieve a competitive performance with 75.21% mAP, and it surpasses all previous models that employ a single-scale feature map for detection. It falls behind the latest model GGHL [19]. We further investigate why such a large margin exists and speculate that GGHL engages the FPN structure which is more robust to scale variation. Besides, GGHL elaborately devises components for joint optimization of multiple subtasks, facilitating a SOTA result. Even so, our model still achieves outstanding performance in detecting densely distributed objects and small objects, such as the SV, LV, SH, and ST categories, and it outperforms GGHL by 2.72%, 3.68%, 2.70%, and 1.00%. We observe that in the GTF and SBF categories, the performance of our model drastically lags behind most methods, which may be attributed to weak features and small amounts of these two categories. Our work mainly focuses on the implementation of adaptive sampling strategy and general OBB representation and pays less attention to designing a complex feature enhancement module, which may be inconducive to weak feature encoding, especially for those categories with weak features that only have a relatively small number of instances.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE COMPARISONS USING DIFFERENT ONE-STAGE ANCHOR-FREE DETECTORS ON THE DOTA1.0 DATASET

Methods	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
IENet [11]	R-101-FPN	57.14	80.20	64.54	39.82	32.07	49.71	65.01	52.58	81.45	44.66	78.51	46.54	56.73	64.40	64.24	57.14
AxisLearning [14]	R-101-FPN	79.53	77.15	38.59	61.15	67.53	70.49	76.30	89.66	79.07	83.53	47.27	61.01	56.28	66.06	36.05	65.98
O^2-DNet [15]	H-104	89.31	82.14	47.33	61.21	71.32	74.03	78.62	90.76	82.23	81.36	60.93	60.17	58.21	66.98	61.03	71.04
P-RSDet [16]	R-101	88.58	77.84	50.44	69.29	71.10	75.79	78.66	90.88	80.10	81.71	57.92	63.03	66.30	69.77	63.13	72.30
BBAVectors [17]	R-101	88.35	79.96	50.69	62.18	78.43	78.98	87.94	90.85	83.58	84.35	54.13	60.24	65.22	64.28	55.70	72.32
DRN [12]	H-104	89.45	83.16	48.98	62.24	70.63	74.25	83.99	90.73	84.60	85.35	55.76	60.79	71.56	68.82	63.92	72.95
TricubeNet [42]	H-104	88.28	80.46	47.32	70.09	76.97	72.97	84.52	90.73	83.87	84.60	56.92	62.91	73.36	68.74	71.36	74.22
GGHL [19]	D-53-FPN	89.74	85.63	44.50	77.48	76.72	80.45	86.16	90.83	88.18	86.25	67.07	69.40	73.38	68.45	70.14	76.95
M2VDet	R-101	86.32	83.91	55.25	58.04	79.44	84.13	88.86	90.85	85.73	87.25	44.26	67.03	73.36	72.88	70.79	75.21

Note: R-101, H-104, D-53 denote ResNet-101, Hourglass-104, Darknet-53, respectively. The performances are evaluated on the testing set through the DOTA online server. The best and second best results are marked in Red and Blue colors.

TABLE II
QUANTITATIVE COMPARISONS USING DIFFERENT ANCHOR-FREE AND ANCHOR-BASED DETECTORS ON THE HRSC2016 DATASET

Methods	Backbone	AF	Image Size	mAP (07)	mAP (12)
IENet [11]	R-101-FPN	✓	1024×1024	75.01	-
AxisLearning [14]	R-101-FPN	✓	800×800	78.15	-
BBAVectors [17]	R-101	✓	608×608	88.60	-
DRN [12]	H-104	✓	768×768	-	92.70
DAL [27]	R-101-FPN	✗	800×800	89.77	-
S^2A-Net [24]	R-101-FPN	✗	800×800	90.17	95.01
OFA-Net [43]	R-50-FPN	✗	800×800	89.80	-
Oriented RepPoints [44]	R-50-FPN	✓	-	90.38	97.26
M2VDet	R-101	✓	800×800	90.18	94.96

Note: R-50, R-101, H-104 denote ResNet-50, ResNet-101, Hourglass-104, respectively. AF indicates anchor-free. mAP (07) and mAP (12) represent the results under VOC2007 and VOC2012 metrics. The best and second best results are marked in Red and Blue colors.

TABLE III
QUANTITATIVE COMPARISONS USING DIFFERENT ANCHOR-FREE AND ANCHOR-BASED DETECTORS ON THE UCAS-AOD DATASET

Methods	Backbone	AF	Image Size	Car	Airplane	mAP
DAL [27]	R-101-FPN	✗	800×800	89.25	90.49	89.87
OFA-Net [43]	R-50-FPN	✗	800×800	89.88	90.54	90.21
P-RSDet [16]	R-101	✓	512×512	87.38	92.69	90.03
Oriented RepPoints [44]	R-50-FPN	✓	-	89.51	90.70	90.11
M2Vdet	R-101	✓	800×800	89.83	90.62	90.23

Note: AF indicates anchor-free. All results are evaluated with VOC2007 metric. The best and second best results are marked in Red and Blue colors.

2) Results on HRSC2016: We then evaluate the performance of our M2VDet on the HRSC2016 dataset. As shown in Table II, we report the results with both VOC2007 and VOC2012 metrics for a comprehensive comparison. Our M2VDet achieves 90.18% and 94.96% mAP under the VOC2007 metric and VOC2012 metric, surpassing several anchor-free and anchor-base models. Without employing the FPN structure that is more robust to scale variations, our model is a little behind the SOTA result achieved by Oriented RepPoints [44], with only 0.2% mAP. The detection result reveals that our M2VDet is effective in detecting objects with excessive aspect ratios and significant scale variations.

3) Results on UCAS-AOD: We go on to make a comparison of our model on the UCAS-AOD dataset. Table III illustrates that our M2VDet outperforms other methods with 90.23% mAP. The high performance once again reveals that M2VDet is superior in detecting small objects.

4) Results on SODA-A: We finally conduct experiments on the SODA-A dataset, and the results of evaluation metrics are listed in Table IV. With ResNet-50 as the backbone, our M2VDet outperforms the existing representative methods with 37.0% AP under the input resolution of 1200 × 1200. Even though the input resolution is 800 × 800, the M2VDet achieves 33.7% AP, surpassing all one-stage detectors and most of the two-stage detectors. This remarkable performance can be attributed to the well-designed modules in the M2VDet. On the one hand, the adaptive IoU threshold can generate more positive samples for small objects, which contributes to the optimization of size-limited objects; on the other hand, the general OBB representation method possesses a stable ability to regress nearly axis-aligned objects, and these types of objects are abundant in this dataset. In addition, there is no rigid limitation for the number of detections, which alleviates missed detections for considerably densely packed scenes. When employing ResNet-101 as the backbone, the model obtains negligible performance gain or decline with 0.1% AP. We hypothesize that a deeper network may do little to cope with objects with limited size and extreme density. We further demonstrate the category-wise results in Table V. It is observed that in the category of helicopter and container, the M2VDet performs way ahead of the comparative models. With the larger input resolution, the performance of detecting small vehicles is definitely improved by 7.1% AP, which indicates that the larger resolution is beneficial to detecting small objects since it can provide more detailed information for such tiny ones.

D. Ablation Study

In this section, we conduct ablation experiments on the DOTA1.0 dataset to evaluate the effectiveness of each proposed component. We train the model for about 80 epochs on the training set and evaluate the performance on the validation set for ablation purposes, following the same implementation details as the pretraining process.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE COMPARISONS USING DIFFERENT ANCHOR-FREE AND ANCHOR-BASED DETECTORS ON THE SODA-A DATASET

Methods	Backbone	AF	AP	AP_{50}	AP_{75}	AP_{eS}	AP_{rS}	AP_{gS}	AP_{N}	Param.	FLOPs
two-stage detectors											
Rotated Faster RCNN* [1]	R-50-FPN	✗	32.5	70.1	24.3	11.9	27.3	42.2	34.4	41.14M	292.25G
RoI Transformer* [22]	R-50-FPN	✗	36.0	73.0	30.1	13.5	30.3	46.1	39.5	55.08M	306.20G
Gliding Vertex* [13]	R-50-FPN	✗	31.7	70.8	22.6	11.7	27.0	41.1	33.8	41.14M	292.25G
Oriented RCNN* [26]	R-50-FPN	✗	34.4	70.7	28.6	12.5	28.6	44.5	36.7	41.13M	292.44G
DoDet* [45]	R-50-FPN	✗	31.6	68.1	23.4	11.3	26.3	41.0	33.5	69.34M	555.49G
one-stage detectors											
Rotated RetinaNet* [3]	R-50-FPN	✗	26.8	63.4	16.2	9.1	22.0	35.4	28.2	36.16M	800.21G
S^2ANet* [24]	R-50-FPN	✗	28.3	69.6	13.1	10.2	22.8	35.8	29.5	38.64M	732.74G
Oriented RepPoints* [44]	R-50-FPN	✓	26.3	58.8	19.0	9.4	22.6	32.4	28.5	55.66M	827.21G
M2Vdet	R-50	✓	33.7	71.2	26.8	9.7	28.5	43.9	37.7	52.81M	282.47G
M2Vdet*	R-50	✓	37.0	75.3	31.4	14.3	31.3	47.1	39.0	52.81M	636.09G
M2Vdet	R-101	✓	33.8	71.2	27.7	9.8	28.5	42.3	37.5	71.75M	329.88G
M2Vdet*	R-101	✓	36.9	75.1	31.3	14.2	30.8	44.9	39.7	71.75M	742.78G

Note: R-50, R-101 denote ResNet-50, ResNet-101, respectively. * indicates the input resolution is 1200×1200 pixels during training and testing. AF indicates anchor-free. The best and second best results are marked in Red and Blue colors.

TABLE V
CATEGORY-WISE AP OF DIFFERENT ANCHOR-FREE AND ANCHOR-BASED DETECTORS ON THE SODA-A DATASET

Methods	AF	airplane	helicopter	s-vehicle	l-vehicle	ship	container	s-tank	s-pool	windmill	AP
Rotated Faster RCNN* [1]	✗	49.4	18.1	33.4	19.6	43.5	29.8	42.8	34.1	21.9	32.5
RoI Transformer* [22]	✗	53.2	21.4	36.1	25.9	46.4	35.7	44.6	36.9	23.5	36.0
Gliding Vertex* [13]	✗	46.7	12.8	33.3	21.9	43.4	29.8	43.3	31.2	22.7	31.7
Oriented RCNN* [26]	✗	52.2	20.2	34.4	24.4	45.2	32.1	43.1	36.3	22.2	34.4
DoDet* [45]	✗	49.4	19.8	32.1	17.3	41.3	26.0	42.2	34.7	21.3	31.6
Rotated RetinaNet* [3]	✗	42.0	16.8	29.9	10.0	35.1	23.7	35.1	30.7	18.1	26.8
S^2ANet* [24]	✗	41.5	20.4	31.2	14.0	36.7	26.1	29.6	33.8	21.6	28.3
Oriented RepPoints* [44]	✓	51.7	8.5	30.3	2.6	28.0	19.6	40.3	33.2	21.9	26.3
M2Vdet	✓	51.0	25.1	31.2	20.6	43.9	38.0	38.1	34.7	20.5	33.7
M2Vdet*	✓	53.9	25.6	38.3	25.3	47.0	41.1	43.7	35.9	22.1	37.0

Note: All models employ ResNet-50 as their backbone. * indicates the input resolution is 1200×1200 pixels during training and testing. AF indicates anchor-free. The small-vehicle, large-vehicle, storage-tank, and swimming pool are abbreviated as s-vehicle, l-vehicle, s-tank, and s-pool, respectively. The best and second best results are marked in Red and Blue colors.

1) Ablation Experiments on the Proposed Components: The ablation results of each proposed component are listed in Table VI. First, we evaluate the effect of positive candidate regions' scope. Without employing the adaptive IoU threshold when generating positive candidate regions, we fix the constant threshold to 0.7 and 0.3 as many classic methods, and we also set the threshold to 0.5 by referring to our adaptive threshold. Then, we get mAP of 68.53%, 70.81%, and 71.73%, with a drop of 4.46%, 2.18%, and 1.26%, respectively, as shown in Table VI. We further exhibit the detection details for each category in Table VII. It can be observed that the adaptive threshold contributes more to objects with weak features, like BR and HC, and objects similar to the background, such as GTF. When setting the constant threshold to 0.7, the AP for small-scale objects reduces precipitously, typical for SV whose AP decreased by 4.25%. We attribute this observation to the intuition that only a few positive locations are assigned for small objects under such a large threshold. As expected, the performance of small objects rises when we choose a small threshold (0.3), and weak objects also attain improvement since sufficient positive samples provide powerful features. Unfortunately, objects for a certain category, among which scales vary extensively, may sustain severe performance degradation under a small threshold, such as harbor with a 2.73% decline in AP. It implies that a smaller threshold may degrade the quality of positive samples, especially for those objects with significant differences in scales. When we set the threshold to a median value of 0.5, the mAP is improved over the former threshold, but it still falls behind the adaptive threshold. Hence, we can conclude that the adaptive IoU threshold is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION EXPERIMENTS ON THE DOTA1.0 DATASET

Image Size	i) Positive Candidate Regions		ii) OBB Representation		iii) Regression Loss		iv) Center Offset Branch	mAP	Model Parameters (MB)	FLOPs (G)	Inference Speed (fps)
	Adaptive Threshold	Constant Threshold	1×20-D vectors	1×12-D vectors	GWD loss	Smooth-L1 loss					
608×608	✓		✓		✓	✓	✓	72.99	71.75	190.58	23.82
608×608		✓($t = 0.7$)	✓		✓	✓	✓	68.53 (-4.46)	71.75	190.58	24.02
608×608		✓($t = 0.3$)	✓		✓	✓	✓	70.81 (-2.18)	71.75	190.58	23.58
608×608		✓($t = 0.5$)	✓		✓	✓	✓	71.73 (-1.26)	71.75	190.58	23.17
608×608	✓			✓		✓	✓	70.40 (-2.59)	71.73	190.15	23.49
608×608	✓			✓	✓	✓	✓	72.36 (-0.63)	71.73	190.15	24.31
608×608	✓		✓			✓	✓	71.01 (-1.98)	71.75	190.58	23.06
608×608	✓		✓		✓	✓		72.61 (-0.38)	71.16	176.92	23.94

Note: The inference speed includes the network inference process and OBB decoding process on an RTX 3090 GPU. The performances are evaluated on the validation set of the DOTA1.0.

TABLE VII
DETAILED PERFORMANCE OF ABLATION EXPERIMENTS ON THE DOTA1.0 DATASET

Methods	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
M2VDet	94.09	75.00	46.97	**59.52**	73.96	**87.09**	**92.57**	94.45	64.21	87.60	66.18	64.93	73.56	**63.12**	**51.61**	**72.99**
Constant $t = 0.7$	93.31	75.48	37.64	53.02	69.71	86.47	90.51	94.23	65.70	67.40	68.10	65.95	72.50	61.96	25.94	68.53
Constant $t = 0.3$	93.81	74.60	43.68	55.23	71.08	85.78	91.84	94.14	61.94	87.46	67.00	64.71	70.83	60.60	39.45	70.81
Constant $t = 0.5$	94.18	74.85	46.06	54.02	**74.35**	86.65	92.52	94.68	65.84	**87.96**	69.53	64.87	73.49	61.92	35.03	71.73
1×12-D w/o GWD loss	**94.46**	75.40	45.06	51.55	69.82	85.48	89.87	94.11	65.21	85.38	63.69	**67.29**	72.12	62.95	33.58	70.40
1×12-D w/ GWD loss	93.94	**79.03**	**47.48**	50.83	71.02	85.75	92.37	94.70	67.30	87.88	**70.51**	64.95	73.03	60.75	45.85	72.36
wo/ GWD loss	93.20	71.73	44.10	58.49	74.20	85.85	90.74	94.78	61.05	86.96	66.63	64.60	**74.17**	60.89	37.72	71.01
wo/ offset branch	94.07	74.84	46.06	51.97	73.72	86.52	92.28	**95.01**	**67.40**	87.68	69.42	67.25	72.32	62.79	47.74	72.61

Note: Bold indicates the best result.

more suitable for generating positive candidate regions of different kinds of objects with variable scales.

Second, we investigate the proposed OBB representation method. Concretely, we analyze the availability of the midpoints-aware vectors' decomposition approach. We retain the standard form of the midpoints-aware vectors and describe the OBB as a 1×12-dimensional (1×12-D) vector instead of a 1×20-dimensional (1×20-D) vector, with four parameters for the circumscribed HBB and eight parameters for the midpoints-aware vectors. As listed in Table VI, the evaluation results when employing 1×12-D representation drop mAP by 0.63%. We observed that the nearly axis-aligned OBB fails to locate accurately under such representation, as illustrated in Fig. 6. By contrast, our proposed OBB representation with a 1×20-D vector is, more general, in describing arbitrary convex quadrilaterals and provides a unified manner to regress axis-aligned and oriented objects completely.

Third, we analyze the loss function for the regression branch. We merely employ smooth-$L1$ loss to supervise the OBB regression branch, including the circumscribed HBB and the midpoint-aware vectors. The mAP metric decreases by 1.98% compared to the combinational loss, as shown in Table VI. We further analyze the detection details in Table VII: objects similar to the background achieve better performance when employing the GWD-based loss since GWD-based loss



(a) (b)

Fig. 6. Illustration of detection results of nearly axis-aligned objects. (a) OBB representation with a 1×12-dimensional vector. (b) OBB representation with a 1×20-dimensional vector.

makes the regression branch easier to optimize, which further drives the feature encoding network to learn more refined features that may benefit the classification branch. Based on this, we prove that GWD-based loss, which acts as an IoU-manner loss, is more effective in supervising the circumscribed HBB regression. It furthermore facilitates the optimization of the midpoint-aware vectors by providing more precise midpoints.

Finally, we remove the center offset branch to simplify the OBB prediction process. In this case, we substitute the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

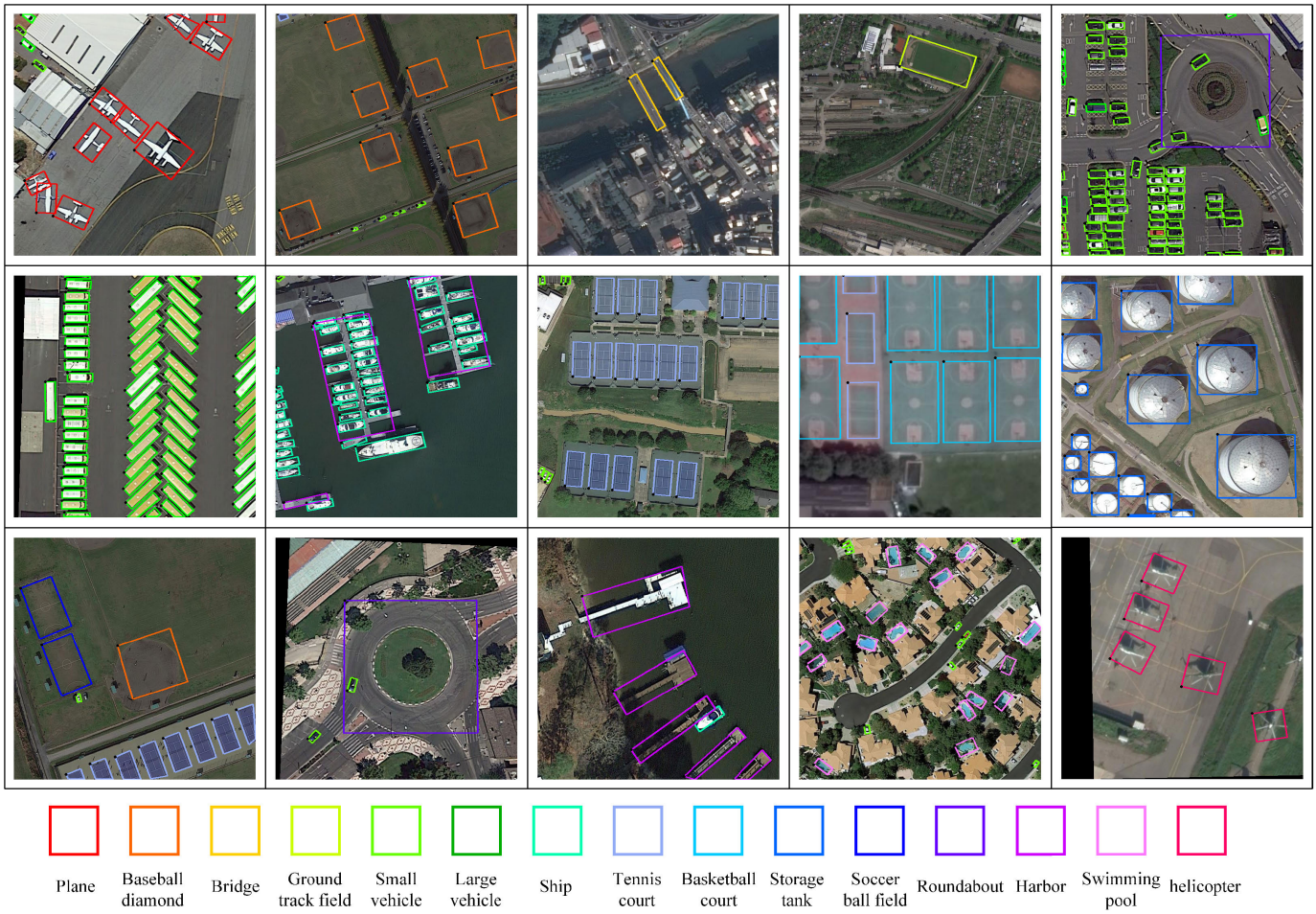


Fig. 7. Representative detection results of the M2VDet on the DOAT1.0 dataset.

integer center point of the circumscribed HBB with the raw floating-point number in the OBB regression branch. As shown in Table VI, the detection mAP drops slightly from 72.99% to 72.61%, while the computational complexity and model parameters reduce dramatically. We further train the model with an input resolution of 800×800 for an additional 50 epochs to verify whether the center offset branch can be discarded for a more concise pipeline. As a result, there is a decrease of 0.88% mAP compared to the M2VDet, which suggests that the center offset branch gains more performance boosts as the input resolution increases. Therefore, a separate branch for the center offset regression is more convenient for optimization since the value of the center offset is relatively small compared to the HBB's parameters.

2) Ablation Experiments on Hyperparameters: Equation (3) introduces a hyperparameter r to generate the adaptive IoU threshold, which is a normalization factor that balances the sensitivity of the IoU thresholds to various scales. We conduct a set of experiments with varying values of r from 0.2 to 1.0 to investigate the effect on the performance. Specifically, we set r to 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, and 1.0 in sequence, and the results are listed in Table VIII. Using the adaptive IoU threshold with different values of r is observed to basically outperform that of using the constant IoU threshold. When varying the hyperparameter r, the performance gap is around 1.5%, and the model achieves the best performance with $r = 0.5$. To be specific, the smaller value of r degenerates the detection performance due to only a small amount of positive samples can be assigned for cross-scale objects under such a strict IoU threshold. As the value of r increases to a certain level, the optimization process is more likely to be dominated by large-scale objects or easy objects, which misleads the optimization preference for cross-scale objects and then damages the detection performance. According to experimental results, the detection performance is relatively robust when varying the value of r from 0.3 to 0.5. We set $r = 0.5$ to pursue a higher performance for all experiments.

TABLE VIII
VARYING r FOR THE ADAPTIVE IOU MAPPING FUNCTION ON THE DOTA1.0 DATASET

r	0.2	0.3	0.4	0.5	0.6	0.8	1.0
mAP	71.80	72.90	72.68	**72.99**	72.31	72.33	71.44

Note: Bold indicates the best result.

In addition, a threshold δ is employed to determine positive samples for the regression branch. We further explore the performance gap when adopting different thresholds, as shown in Table IX. It is revealed that too high or too low a threshold could damage performance. When adopting the threshold of 1.0, only Gaussian peak points are utilized to regress

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.



Fig. 8. Representative detection results of the M2VDet on the HRSC2016 dataset.



Fig. 9. Representative detection results of the M2VDet on the UCAS-AOD dataset.

TABLE IX

VARYING δ TO DETERMINE POSITIVE SAMPLES FOR THE REGRESSION BRANCH ON THE DOTA1.0 DATASET

δ	0	0.1	0.2	0.4	0.6	0.7	0.8	1.0
mAP	71.32	72.32	72.81	72.70	**72.99**	72.04	71.66	71.25

Note: Bold indicates the best result.

OBBs, then we obtain a typical keypoint-based model. The model suffers from performance deterioration since the drift of peak points considerably impacts the location accuracy. As the threshold decreases, more locations surrounding the peak points are assigned as positive samples, which lessens the requirement for extracting the peak points precisely and naturally embraces performance gain. The performance drops significantly when the threshold value is sufficiently small, such as in the extreme case with the threshold equal to 0. We argue that a smaller threshold retains enough samples for regression, while the bounding box at the location away from the peak point is of lower quality, which raises the difficulty of optimizing the regression branch and biases the training process toward optimizing the OBBs of those hard samples. With a wide range of 0.2–0.6, the threshold δ has little impact on mAP. We set $\delta = 0.6$ to guarantee that the training process prioritizes objects with high-quality locations for all experiments.

E. Qualitative Results

The representative detection results are exhibited on four datasets: Fig. 7 for the DOTA1.0 dataset, Fig. 8 for the HRSC2016 dataset, Fig. 9 for the UCAS-AOD dataset, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

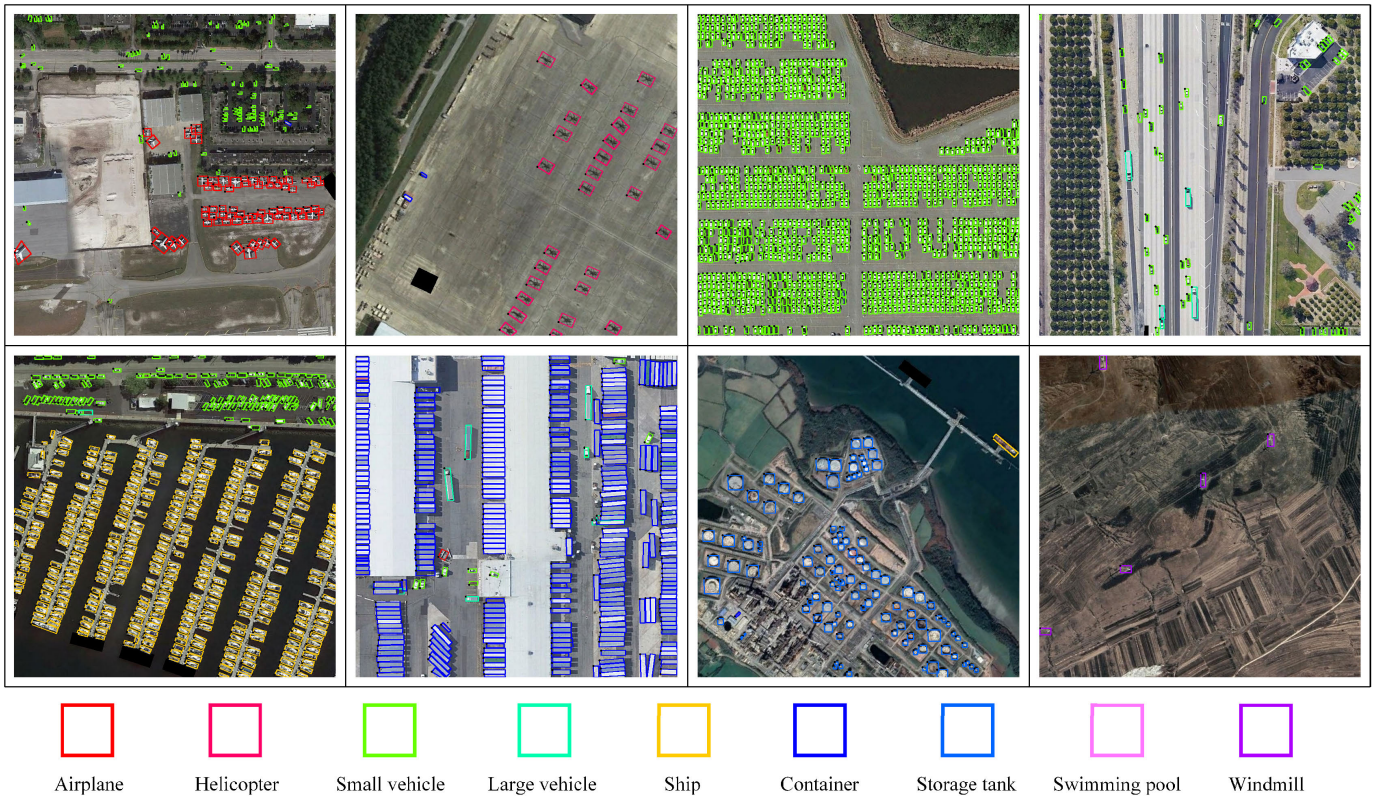


Fig. 10. Representative detection results of the M2VDet on the SODA-A dataset.

Fig. 10 for the SODA-A dataset. From the visualization results, we can conclude the following advantages of our method: 1) the proposed general OBB representation can accurately locate nearly axis-aligned objects and other oriented objects through one branch, even if the objects are in dense distribution, typified by the small-vehicle category in Figs. 7, 9, and 10, as well as the container category in Fig. 10; 2) the proposed adaptive sampling strategy enables the M2VDet to perform satisfactorily in detecting small objects with arbitrary orientation. As illustrated in Fig. 10, the recall in detecting small objects is relatively high; and 3) without utilizing the FPN structure, the M2VDet is still effective in detecting objects with excessive aspect ratios and significant scale variations, as shown in Fig. 8, but it relies on a superior pretrained weight or sufficient training data. Our work adopts the pretrained weight of the backbone and FFM from the DOTA dataset. We have attempted to train the M2VDet on the HRSC2016 dataset based on the pretrained backbone from the ImageNet dataset, but we just acquired a poor performance. The reason may be that the high-resolution single-scale feature map is inherently inadequate to encode objects with extreme scale variations, and the proposed OBB representation method requests a certain number of training samples for better optimization.

V. Conclusion

In this article, we present an effective one-stage anchor-free oriented object detector, called M2VDet, to detect remote-sensing objects with arbitrary orientations and diverse scales. Our work mainly focuses on studying the adaptive sampling strategy and the general OBB representation method based on a simple CNN architecture. Specifically, an adaptive sampling strategy is proposed to generate positive sample spaces, and the 2-D-oriented Gaussian distribution is then employed for weighting the sample spaces. As a result, the positive candidate regions are constructed that are more in line with the characteristics of oriented objects. Besides, a general OBB representation method with the midpoints-to-vertices strategy is designed to describe objects shaped with arbitrary convex quadrilaterals, which performs more promising in capturing nearly axis-aligned objects and other oriented objects in a unified pattern. Extensive experiments on four public datasets indicate that the proposed M2VDet possesses superior ability in detecting oriented objects in remote-sensing images, especially for densely packed objects and small objects with arbitrary orientation.

However, performance degradation still exists compared to current SOTA methods, especially on the DOTA1.0 dataset. We clarify that the adaptive sampling strategy and the general OBB representation of anchor-free oriented object detectors are the main components of our method. There is no attempt to make the model's structure more complex, which might lead to features being lacking for faint objects. Therefore, it is unlikely that the detection performance of faint objects would increase. Besides, our M2VDet employs the high-resolution feature map with a single scale to detect objects, which is inherently insufficient to encode objects with extensive scales and may result in low precision when locating large objects, especially in cases where the instance number is relatively small. These issues are still to be studied in the future.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[2] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[3] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020, doi: 10.1109/TPAMI.2018.2858826.

[4] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[5] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[6] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[7] X. Zhou, J. Zhuo, and P. Krähenbühl, "Bottom-up object detection by grouping extreme and center points," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 850–859.

[8] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9656–9665.

[9] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.

[10] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[11] Y. Lin, P. Feng, J. Guan, W. Wang, and J. Chambers, "IENet: Interacting embranchment one stage anchor free detector for orientation aerial object detection," 2019, *arXiv:1912.00969*.

[12] X. Pan et al., "Dynamic refinement network for oriented and densely packed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11204–11213.

[13] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021, doi: 10.1109/TPAMI.2020.2974745.

[14] Z. Xiao, L. Qian, W. Shao, X. Tan, and K. Wang, "Axis learning for orientated objects detection in aerial images," *Remote Sens.*, vol. 12, no. 6, p. 908, Mar. 2020.

[15] H. Wei, Y. Zhang, Z. Chang, H. Li, H. Wang, and X. Sun, "Oriented objects as pairs of middle lines," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 268–279, Nov. 2020.

[16] L. Zhou, H. Wei, H. Li, W. Zhao, Y. Zhang, and Y. Zhang, "Arbitrary-oriented object detection in remote sensing images based on polar coordinates," *IEEE Access*, vol. 8, pp. 223373–223384, 2020.

[17] J. Yi, P. Wu, B. Liu, Q. Huang, H. Qu, and D. Metaxas, "Oriented object detection in aerial images with box boundary-aware vectors," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 2149–2158.

[18] S. Lang, F. Ventola, and K. Kersting, "DAFNe: A one-stage anchor-free approach for oriented object detection," 2021, *arXiv:2109.06148*.

[19] Z. Huang, W. Li, X.-G. Xia, and R. Tao, "A general Gaussian heatmap label assignment for arbitrary-oriented object detection," *IEEE Trans. Image Process.*, vol. 31, pp. 1895–1910, 2022.

[20] X. Yang et al., "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 8232–8241.

[21] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," 2019, *arXiv:1908.05612*.

[22] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2844–2853.

[23] X. Yang, J. Yan, W. Liao, X. Yang, J. Tang, and T. He, "SCRDet++: Detecting small, cluttered and rotated objects via instance-level feature denoising and rotation loss smoothing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2384–2399, Feb. 2023, doi: 10.1109/TPAMI.2022.3166956.

[24] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511.

[25] J. Han, J. Ding, N. Xue, and G.-S. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2785–2794.

[26] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Oct. 2021, pp. 3500–3509.

[27] Q. Ming, Z. Zhou, L. Miao, H. Zhang, and L. Li, "Dynamic anchor learning for arbitrary oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 2355–2363.

[28] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.

[29] J. Wang, C. Xu, W. Yang, and L. Yu, "A normalized Gaussian Wasserstein distance for tiny object detection," 2021, *arXiv:2110.13389*.

[30] X. Zhang, F. Wan, C. Liu, R. Ji, and Q. Ye, "FreeAnchor: Learning to match anchors for visual object detection," in *Proc. Neural Inf. Process. Syst.*, 2019, pp. 147–155.

[31] B. Zhu et al., "AutoAssign: Differentiable label assignment for dense object detection," 2020, *arXiv:2007.03496*.

[32] K. Kim and H. S. Lee, "Probabilistic anchor assignment with IoU prediction for object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 355–371.

[33] L. Hou, K. Lu, J. Xue, and Y. Li, "Shape-adaptive selection and measurement for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 923–932.

[34] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 11830–11842.

[35] W. Qian, X. Yang, S. Peng, Y. Guo, and C. Yan, "Learning modulated loss for rotated object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 2458–2466.

[36] Z. Chen et al., "PIoU loss: Towards accurate oriented object detection in complex environments," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 195–211.

[37] X. Yang et al., "Learning high-precision bounding box for rotated object detection via Kullback–Leibler divergence," in *Proc. Neural Inf. Process. Syst.*, 2021, pp. 18381–18394.

[38] X. Yang et al., "The KFIoU loss for rotated object detection," 2022, *arXiv:2201.12558*.

[39] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. Int. Conf. Pattern Recognit. Appl. Methods*, vol. 1, 2017, pp. 324–331.

[40] H. Zhu, X. Chen, W. Dai, K. Fu, Q. Ye, and J. Jiao, "Orientation robust object detection in aerial images using deep convolutional neural network," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 3735–3739.

[41] G. Cheng et al., "Towards large-scale small object detection: Survey and benchmarks," 2022, *arXiv:2207.14096*.

[42] B. Kim, J. Lee, S. Lee, D. Kim, and J. Kim, "TricubeNet: 2D kernel-based object representation for weakly-occluded oriented object detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 3421–3430.

[43] M. Zou, Z. Hu, Y. Guan, Z. Gan, C. Guan, and S. Leng, "Feature adaption with predicted boxes for oriented object detection in aerial images," in *Proc. Pacific Rim Int. Conf. Artif. Intell.*, vol. 13033, 2021, pp. 366–378.

[44] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented reppoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1829–1838.

[45] G. Cheng et al., "Dual-aligned oriented detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618111.



Xueru Xu (Student Member, IEEE) received the B.S. and M.S. degrees from Chongqing University, Chongqing, China, in 2016 and 2019, respectively. She is currently pursuing the Ph.D. degree with the Huazhong University of Science and Technology, Wuhan, China.

Her research interests include object detection and fine-grained recognition in remote-sensing applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.



Zhong Chen received the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 2006.

He is currently an Associate Professor with the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, China. His research interests include computer vision and deep learning, with a focus on remote-sensing image processing.



Guoyou Wang received the M.S. degree from the Huazhong University of Science and Technology, Wuhan, China, in 1992.

He is currently a Professor with the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology. His research interests include computer vision and neural network interpretability.



He Deng received the Ph.D. degree in control science and engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2011.

He is currently a Professor with the Wuhan University of Science and Technology, Wuhan. His research interests are artificial intelligence and medical image processing.



Longji Yu received the B.S. degree from the Huazhong University of Science and Technology, Wuhan, China, in 2020, where he is currently pursuing the M.S. degree in artificial intelligence.

His research interests include object detection and object tracking.



Qimeng Chen received the B.S. degree from the Naval University of Engineering, Wuhan, China, in 2010, and the M.S. degree from the Huazhong University of Science and Technology, Wuhan, in 2023.

His research interests include computer vision and remote-sensing image interpretation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:23:14 UTC from IEEE Xplore. Restrictions apply.